# Project Design Phase-I

**Proposed Solution Template**

**Date:** 24 February 2026
**Team ID:** 699ee17ef292113d819d8e62
**Project Name**:Harnessing Machine Learning for Blood Pressure Analysis
**Maximum Marks:** 2 Marks

**Proposed Solution Template:**
Project team shall fill the following information in proposed solution template.

| S.No. | Parameter | Description |
|---|---|---|
| 1 | Problem Statement (Problem to be solved) | Hypertension is one of the leading causes of cardiovascular diseases and often remains undetected until severe complications occur. Many people do not regularly monitor their blood pressure or understand their risk level. The problem is to develop a system that can predict and classify stages of hypertension using clinical parameters and lifestyle data to support early detection and monitoring. |
| 2 | Idea / Solution Description | The proposed system uses machine learning algorithms to analyze patient health data such as blood pressure readings, age, lifestyle habits, and medical history. The system processes this data to predict the stage of hypertension and provide risk assessment. It also suggests lifestyle recommendations such as diet changes, exercise, and monitoring habits to help users manage their health effectively. |
| 3 | Novelty / Uniqueness | The uniqueness of the proposed solution lies in combining clinical parameters and lifestyle factors to create an intelligent hypertension prediction model. Unlike traditional monitoring systems, this approach uses machine learning techniques to provide early risk prediction and personalized recommendations for preventive healthcare. |
| 4 | Social Impact / Customer Satisfaction | The system helps individuals monitor their hypertension risk and encourages early medical attention. It can improve public health awareness, reduce healthcare costs through early detection, and assist healthcare professionals in decision support. This leads to improved patient satisfaction and better management of hypertension-related health risks. |
| 5 | Business Model (Revenue Model) | The solution can be deployed as a healthcare analytics platform integrated with hospitals, clinics, or wearable health devices. Revenue can be generated through subscription-based health monitoring services, partnerships with healthcare institutions, and licensing the predictive analytics model to medical organizations. |
| 6 | Scalability of the Solution | The system can be scaled to support large healthcare datasets and integrated with mobile applications, hospital information systems, and wearable health monitoring devices. Cloud-based deployment allows the solution to handle large numbers of users and extend the model to predict other cardiovascular or lifestyle-related diseases. |